Math 104 - Spring 2026 - Exploration 4.2: Have a Nice Trip

Recent studies in biomechanics and gerontology focused on how individuals respond to a large postural disturbance like tripping over an object. Suppose a group of researchers wants to compare two strategies subjects might use to recover from tripping:

- Lowering - quickly stepping down with front leg and then raising back leg over object
- Elevating - lifting front leg over object

Subjects are trained on one of the two strategies. The researchers harness the subjects for safety and then induce the subject to trip while walking using a concealed mechanical obstacle. Subjects are asked to apply their learned strategy when they feel themselves tripping.

Suppose 24 subjects have agreed to participate in the study. Their heights range from 61 inches to 74 inches. There are 6 individuals whose height was 72 inches (6 feet) or more. For simplicity, we'll call the participants who are 72 inches or taller "tall" and the rest of the group "not tall."

(1) From the point of view of study design, why would it not be reasonable to assign the 6 "tall" people the elevating strategy and the 18 "not tall" people the lowering strategy?

(2) Now suppose you assign participants to the lowering group and the elevating group in such a way that the groups have the same average height. If you saw a difference between the proportion of trips in the two groups, could it be due to the height of the subject? Could it be due to other variables besides the recovery strategy?

Let's explore the process of random assignment.

(3) Open the Randomizing Subjects applet. Next to **Choose variables**, select "height" and press **Randomize**.

What is the average height of people assigned to Group 1?

What is the average height of people assigned to Group 2?

What is the difference between the two averages?

(4) Press **Randomize** again. What is the difference between the two averages this time?

(5) Change the number of repititions from 1 to 198 (to get a total of 200), uncheck Animate, and press **Randomize**. The dot plot displays the difference between the groups' average heights for each of the 200 repetitions. Where is the distribution centered?

(6) Does random assignment *always* balance the heights between the groups equally?
Is there a tendency for the average heights of the two groups to be similar?

Prior research has shown that the likelihood of falling is related to variables such as walking speed and stride rate so we would like the random assignment to distribute these variables equally between the groups as well. What's more, we would like random assignment to balance potential confounding variables *we aren't even aware of.*

(7) Suppose there is a gene for balance that is related to people's ability to recover from a trip. We didn't know about this gene ahead of time, but it is a potential confounding variable. Select the **Reveal gene?** button and choose **gene** from the pull-down menu. The applet now shows whether each subject has the gene and how the proportions of people with the gene differ between the two groups. Does the gene variable tend to equalize between the two groups in the long run? Explain.

(8) Suppose there were other variables that we didn't think to measure. These are also potential confounding variables. Select **Reveal both?** and use the pull-down menu to display the results for subjects' BMI. Does random assignment generally succeed in equalizing this variable between the two groups, or is there a tendency for one group to always have higher BMI? Explain.

(9) Suppose the study finds a statistically significant difference between the two groups. What conclusion would you draw?

To what population would your conclusion apply?